CLASS 8 — SCIENCE · FORMULA SHEET

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | Quick revision, one page

Terms

- Solution: homogeneous mixture of solute and solvent. Solute: the substance dissolved; solvent: the substance in which it dissolves.
- Mass of solution = mass of solute + mass of solvent.
- Water is called the universal solvent as it dissolves a very large number of substances (not all).

Concentration and solubility

- $extMasspercentage = \frac{extmassofsolute}{extmassofsolution} imes 100$
- Saturated solution: cannot dissolve more solute at that temperature. Unsaturated: can dissolve more.
- Solubility: maximum mass of solute that dissolves in 100 g of solvent at a given temperature.

Factors affecting dissolving

- Solubility of most solids in water increases with temperature.
- Solubility of gases in water decreases as temperature rises and increases with pressure.
- Stirring, heating and powdering increase the rate of dissolving.

Solutions, suspensions and colloids

- Solution: very small particles; clear; does not settle; passes through filter paper; no Tyndall effect.
- Suspension (muddy water): large particles; settle on standing; can be filtered; scatter light.
- Colloid (milk): intermediate particles; do not settle; show the Tyndall effect.

MASTERMIND